then add 10 or add 5 to 10 and multiply the result by 3. When we consult the order of operations, however, we find that multiplication always takes precedence over addition regardless of the order they appear in the expression; so the correct answer would be 25.

Alternatively, if we wanted 45 we could add parentheses to the expression making it now `3 * (5 + 10)` to modify the order in which the expression is evaluated. In this case, parenthesis take precedence over every other operator, so 5 is added to 10 and that sum is then multiplied by 3 for the modified result of 45. Parenthesis may also be added to an expression not necessarily to modify the order of operations, but rather just to make things easier to understand. Take the following examples:

```
x = 5 * 5 + 5;    // assigns 30 to x
y = (5 * 5) + 5;  // assigns 30 to y
z = 5 * (5 + 5);  // assigns 50 to z
```

In the first line, the order of operations is followed to result in a final value of 30, which happens to be the same value for the second line because the parenthesis are only their aesthetically. The final line modifies the normal order of operations by bracketing `5 + 5` with parenthesis so that that part of the expression will take precedence. Here, the sum of 5 and 5 is then multiplied by 5 for a total value of 50.

Table 3-3 shows the normal order of operations ranked by highest precedence at the top and lowest at the bottom. We have covered some of these operators in this chapter, while others will be discussed later.

Table 3-3. *Order of Operations*

Symbol	Description	Examples
( )	Parenthesis	(x+y)*z
++ --	Increment, decrement	x++, y--
* / %	Multiplication, division, modulus	x * 1.5
+ -	Addition, subtraction	y + 10
< > <= >=	Less than, greater than comparisons	if (x < 255)
== !=	Is, is not equal to	if (x == HIGH)
&&	Logical AND	if (x == HIGH && x > y)
||	Logical OR	if (x == HIGH || x > y)
= += -= *= /=	Assignment and compound assignments	x = y

Consulting the chart, you can see that division will always happen before addition and subtraction will happen before assignment. There's also some other things thrown in there, but we will get to these before too long.